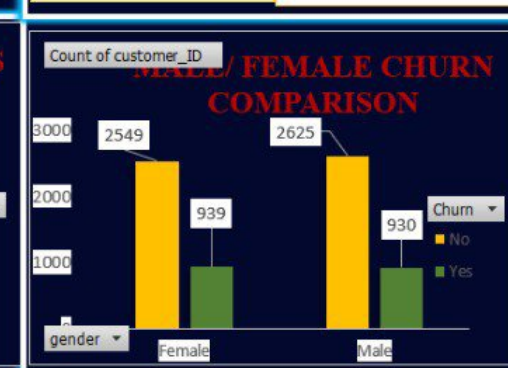
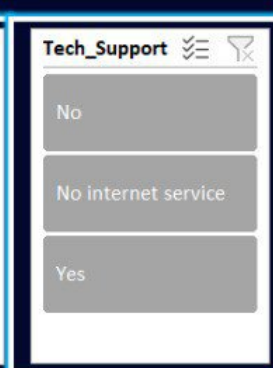
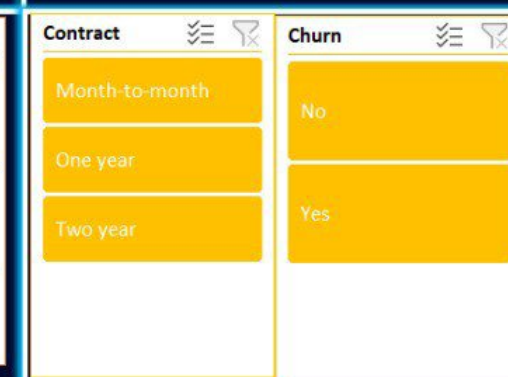
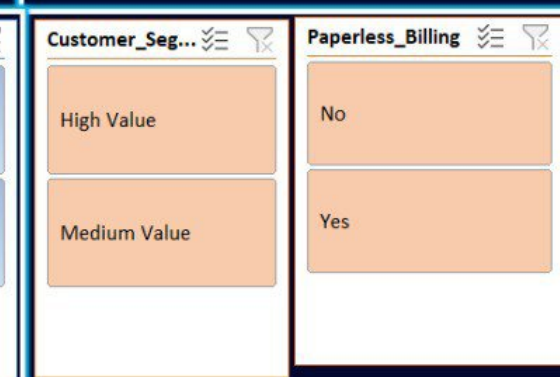
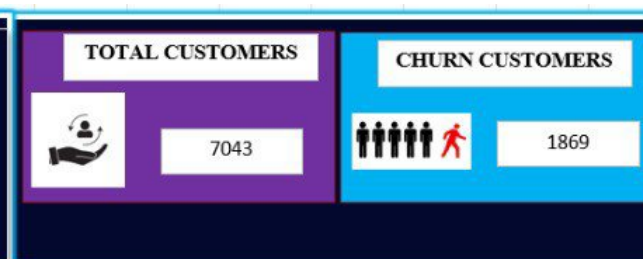
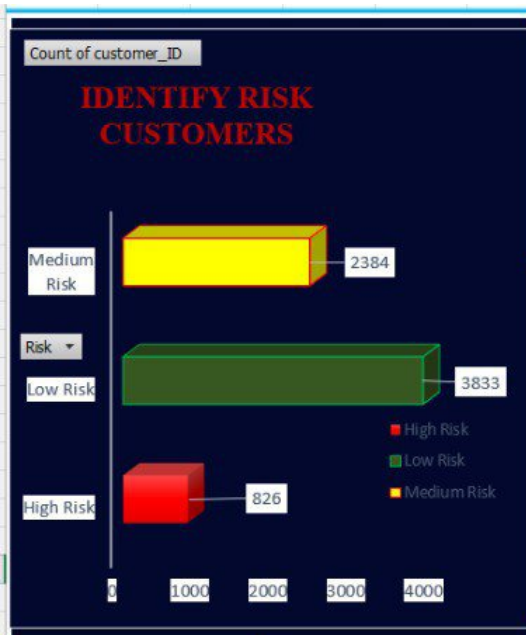


	R	S	T	U	V	W	X	Y	Z	AA	AB	AC	AD	AE	AF
1	Payment Method	Monthly Chrg	Total Chrg	Churn	Customer Stat	CLV	Risk	Customer Segme	Possible Churn Reas	SeniorCitizen Typ	Contract Risk	Internet_Risk	Risk_Score	Churn_Score	CustomerSegment_Score
2	Electronic check	29.85	29.85	No	Active Customer	29.85	Medium Risk	Medium Value	Short Contract	Normal Customer	High churn probability	Normal Charges	2	0	2
3	Mailed check	56.95	1889.5	No	Active Customer	1889.3	Low Risk	Medium Value	No Tech support	Normal Customer	Stable Customer	Normal Charges	1	0	2
4	Mailed check	53.85	108.15	No	Left Customer	107.7	Medium Risk	Medium Value	Short Contract	Normal Customer	High churn probability	Normal Charges	2	1	2
5	Bank transfer (automatic)	82.3	1840.75	No	Active Customer	1903.5	Low Risk	Medium Value	Other	Normal Customer	Stable Customer	Normal Charges	1	0	2
6	Electronic check	70.3	151.65	No	Left Customer	141.4	High Risk	Medium Value	Short Contract	Normal Customer	High churn probability	High Charges	3	1	2
7	Electronic check	99.65	820.5	No	Left Customer	797.2	High Risk	Medium Value	Short Contract	Normal Customer	High churn probability	High Charges	3	1	2
8	Credit card (automatic)	89.1	1949.4	No	Active Customer	1960.2	Medium Risk	Medium Value	Short Contract	Normal Customer	High churn probability	High Charges	2	0	2
9	Mailed check	29.75	301.9	No	Active Customer	297.5	Medium Risk	Medium Value	Short Contract	Normal Customer	High churn probability	Normal Charges	2	0	2
10	Electronic check	104.8	3046.05	No	Left Customer	2934.4	Low Risk	Medium Value	Short Contract	Normal Customer	High churn probability	High Charges	1	1	2
11	Bank transfer (automatic)	58.15	3487.95	No	Active Customer	3481.3	Low Risk	Medium Value	No Tech support	Normal Customer	Stable Customer	Normal Charges	1	0	2
12	Mailed check	49.95	587.45	No	Active Customer	649.35	Medium Risk	Medium Value	Short Contract	Normal Customer	High churn probability	Normal Charges	2	0	2
13	Credit card (automatic)	18.95	326.8	No	Active Customer	303.2	Medium Risk	Medium Value	Other	Normal Customer	Stable Customer	Normal Charges	2	0	2
14	Credit card (automatic)	100.35	5681.1	No	Active Customer	5820.3	Low Risk	High Value	No Tech support	Normal Customer	Stable Customer	High Charges	1	0	3
15	Bank transfer (automatic)	103.7	5036.3	No	Left Customer	5081.3	Low Risk	High Value	Short Contract	Normal Customer	High churn probability	High Charges	1	1	3
16	Electronic check	105.5	2686.05	No	Active Customer	2637.5	Low Risk	Medium Value	Short Contract	Normal Customer	High churn probability	High Charges	1	0	2
17	Credit card (automatic)	113.25	7895.15	No	Active Customer	7814.25	Low Risk	High Value	Other	Normal Customer	Stable Customer	High Charges	1	0	3
18	Mailed check	20.65	1022.95	No	Active Customer	1073.8	Low Risk	Medium Value	Other	Normal Customer	Stable Customer	Normal Charges	1	0	2
19	Bank transfer (automatic)	106.7	7382.25	No	Active Customer	7575.7	Low Risk	High Value	No Tech support	Normal Customer	Stable Customer	High Charges	1	0	3
20	Credit card (automatic)	55.2	528.35	No	Left Customer	552	Medium Risk	Medium Value	Short Contract	Normal Customer	High churn probability	Normal Charges	2	1	2
21	Electronic check	90.05	1862.9	No	Active Customer	1891.05	Medium Risk	Medium Value	Short Contract	Normal Customer	High churn probability	High Charges	2	0	2
22	Electronic check	39.65	39.65	No	Left Customer	39.65	Medium Risk	Medium Value	Short Contract	Senior Citizen	High churn probability	Normal Charges	2	1	2
23	Bank transfer (automatic)	19.8	202.25	No	Active Customer	237.6	Medium Risk	Medium Value	Other	Normal Customer	Stable Customer	Normal Charges	2	0	2
24	Mailed check	20.15	20.15	No	Left Customer	20.15	Medium Risk	Medium Value	Short Contract	Normal Customer	High churn probability	Normal Charges	2	1	2
25	Credit card (automatic)	59.9	3505.1	No	Active Customer	3474.2	Low Risk	Medium Value	Other	Normal Customer	Stable Customer	Normal Charges	1	0	2
26	Credit card (automatic)	59.6	2970.3	No	Active Customer	2920.4	Low Risk	Medium Value	Short Contract	Normal Customer	High churn probability	Normal Charges	1	0	2
27	Bank transfer (automatic)	55.3	1530.6	No	Active Customer	1659	Low Risk	Medium Value	Short Contract	Normal Customer	High churn probability	Normal Charges	1	0	2
28	Electronic check	99.35	4749.15	No	Left Customer	4669.45	Low Risk	Medium Value	Short Contract	Normal Customer	High churn probability	High Charges	1	1	2
29	Electronic check	30.2	30.2	No	Left Customer	30.2	Medium Risk	Medium Value	Short Contract	Normal Customer	High churn probability	Normal Charges	2	1	2
30	Credit card (automatic)	90.25	6369.45	No	Active Customer	6498	Low Risk	High Value	Other	Normal Customer	Stable Customer	Normal Charges	1	0	3
31	Mailed check	64.7	1093.1	No	Left Customer	1099.9	Medium Risk	Medium Value	Short Contract	Normal Customer	High churn probability	Normal Charges	2	1	2
32	Credit card (automatic)	96.35	6766.95	No	Active Customer	6840.85	Low Risk	High Value	Other	Senior Citizen	Stable Customer	High Charges	1	0	3
33	Credit card (automatic)	95.5	181.65	No	Active Customer	191	High Risk	Medium Value	Short Contract	Senior Citizen	High churn probability	High Charges	3	0	2
34	Mailed check	66.15	1874.45	No	Active Customer	1786.05	Low Risk	Medium Value	Other	Normal Customer	Stable Customer	Normal Charges	1	0	2
35	Bank transfer (automatic)	20.2	20.2	No	Active Customer	20.2	Medium Risk	Medium Value	Short Contract	Normal Customer	High churn probability	Normal Charges	2	0	2
36	Bank transfer (automatic)	45.25	45.25	No	Active Customer	45.25	Medium Risk	Medium Value	Short Contract	Senior Citizen	High churn probability	Normal Charges	2	0	2
37	Bank transfer (automatic)	99.9	7251.7	No	Active Customer	7192.8	Low Risk	High Value	Other	Normal Customer	Stable Customer	High Charges	1	0	3
38	Electronic check	69.7	316.9	No	Left Customer	348.5	Medium Risk	Medium Value	Short Contract	Normal Customer	High churn probability	High Charges	2	1	2
39	Credit card (automatic)	74.8	3548.3	No	Active Customer	3440.8	Low Risk	Medium Value	Short Contract	Normal Customer	High churn probability	High Charges	1	0	2
40	Electronic check	106.35	3549.25	No	Left Customer	3615.9	Low Risk	Medium Value	Short Contract	Normal Customer	High churn probability	High Charges	1	1	2
41	Bank transfer (automatic)	97.85	1105.4	No	Left Customer	1076.35	High Risk	Medium Value	Short Contract	Normal Customer	High churn probability	High Charges	3	1	2
42	Mailed check	49.55	475.7	No	Active Customer	495.5	Medium Risk	Medium Value	No Tech support	Normal Customer	Stable Customer	Normal Charges	2	0	2
43	Credit card (automatic)	69.2	4872.35	No	Active Customer	4844	Low Risk	Medium Value	No Tech support	Normal Customer	Stable Customer	Normal Charges	1	0	2
44	Mailed check	20.75	418.25	No	Active Customer	352.75	Medium Risk	Medium Value	Other	Normal Customer	Stable Customer	Normal Charges	2	0	2
45	Credit card (automatic)	79.85	4861.45	No	Active Customer	5030.55	Low Risk	High Value	Other	Normal Customer	Stable Customer	Normal Charges	1	0	3
46	Electronic check	76.3	981.45	No	Active Customer	990.6	Medium Risk	Medium Value	Short Contract	Normal Customer	High churn probability	Normal Charges	2	0	2
47	Electronic check	84.5	3906.7	No	Active Customer	4140.5	Low Risk	Medium Value	Short Contract	Normal Customer	High churn probability	High Charges	1	0	2
48	Mailed check	49.25	97	No	Active Customer	98.5	Medium Risk	Medium Value	Short Contract	Normal Customer	High churn probability	Normal Charges	2	0	2
49	Electronic check	80.65	144.15	No	Left Customer	161.3	High Risk	Medium Value	Short Contract	Normal Customer	High churn probability	High Charges	3	1	2

	A		B		C		D		E		F		G		H		I		J		K		L		M		N		O		P		Q		
1	customer	gender	Senior	Citiz	Partner	Dependent	tenant	Phone Serv	Multiple Lin	Internet Serv	Online Secur	Online Back	Device Protect	Tech Suppor	Streaming TV	Streaming Mov	Contract	Paperless Bill	Paym																
2	7590-VHVEG	Female	0		Yes	No	1	No	No phone service	DSL	No	Yes	No	No	No	No	Month-to-month	Yes	Elec																
3	5575-GNVDE	Male	0		No	No	34	Yes	No	DSL	Yes	No	Yes	No	No	No	One year	No	M																
4	3668-QPYBK	Male	0		No	No	2	Yes	No	DSL	Yes	Yes	No	No	No	No	Month-to-month	Yes	M																
5	7795-CFOCW	Male	0		No	No	45	No	No phone service	DSL	Yes	No	Yes	Yes	No	No	One year	No	Bank tra																
6	9237-HQITU	Female	0		No	No	2	Yes	No	Fiber optic	No	No	No	No	No	No	Month-to-month	Yes	Elec																
7	9305-CDSKC	Female	0		No	No	8	Yes	Yes	Fiber optic	No	No	Yes	No	Yes	Yes	Month-to-month	Yes	Elec																
8	1452-KIOVK	Male	0		No	Yes	22	Yes	Yes	Fiber optic	No	Yes	No	No	Yes	No	Month-to-month	Yes	Credit																
9	6713-OKOMC	Female	0		No	No	10	No	No phone service	DSL	Yes	No	No	No	No	No	Month-to-month	No	M																
10	7892-POOKP	Female	0		Yes	No	28	Yes	Yes	Fiber optic	No	No	Yes	Yes	Yes	Yes	Month-to-month	Yes	Elec																
11	6388-TABGU	Male	0		No	Yes	62	Yes	No	DSL	Yes	Yes	No	No	No	No	One year	No	Bank tra																
12	9763-GRSKD	Male	0		Yes	Yes	13	Yes	No	DSL	Yes	No	No	No	No	No	Month-to-month	Yes	M																
13	7469-LKBCI	Male	0		No	No	16	Yes	No	No	No internet service	No internet service	No internet service	No internet service	No internet service	No internet service	Two year	No	Credit																
14	8091-TTVAX	Male	0		Yes	No	38	Yes	Yes	Fiber optic	No	No	Yes	No	Yes	Yes	One year	No	Credit																
15	0280-XJGEX	Male	0		No	No	49	Yes	Yes	Fiber optic	No	Yes	Yes	No	Yes	Yes	Month-to-month	Yes	Bank tra																
16	5129-JLPIS	Male	0		No	No	25	Yes	No	Fiber optic	Yes	No	Yes	Yes	Yes	Yes	Month-to-month	Yes	Elec																
17	3655-SNQYZ	Female	0		Yes	Yes	69	Yes	Yes	Fiber optic	Yes	Yes	Yes	Yes	Yes	Yes	Two year	No	Credit																
18	8191-XWSZG	Female	0		No	No	52	Yes	No	No	No internet service	No internet service	No internet service	No internet service	No internet service	No internet service	One year	No	M																
19	9959-WOFTK	Male	0		No	Yes	71	Yes	Yes	Fiber optic	Yes	No	Yes	No	Yes	Yes	Two year	No	Bank tra																
20	4190-MFLUW	Female	0		Yes	Yes	10	Yes	No	DSL	No	No	Yes	Yes	No	No	Month-to-month	No	Credit																
21	4183-MYFRB	Female	0		No	No	21	Yes	No	Fiber optic	No	Yes	Yes	No	No	Yes	Month-to-month	Yes	Elec																
22	8779-QRDMV	Male	1		No	No	1	No	No phone service	DSL	No	No	Yes	No	No	Yes	Month-to-month	Yes	Elec																
23	1680-UDCWV	Male	0		Yes	No	12	Yes	No	No	No internet service	No internet service	No internet service	No internet service	No internet service	No internet service	One year	No	Bank tra																
24	1066-JKSGK	Male	0		No	No	1	Yes	No	No	No internet service	No internet service	No internet service	No internet service	No internet service	No internet service	Month-to-month	No	M																
25	3638-WEABW	Female	0		Yes	No	38	Yes	Yes	DSL	No	Yes	No	Yes	No	No	Two year	Yes	Credit																
26	6322-HRPPA	Male	0		Yes	Yes	49	Yes	No	DSL	Yes	Yes	No	Yes	No	No	Month-to-month	No	Credit																
27	6865-JZKNO	Female	0		No	No	30	Yes	No	DSL	Yes	Yes	No	No	No	No	Month-to-month	Yes	Bank tra																
28	6467-CHFZW	Male	0		Yes	Yes	47	Yes	Yes	Fiber optic	No	Yes	No	No	Yes	Yes	Month-to-month	Yes	Elec																
29	8665-UTDZH	Male	0		Yes	Yes	1	No	No phone service	DSL	No	Yes	No	No	No	No	Month-to-month	No	Elec																
30	5248-YGIUN	Male	0		Yes	No	72	Yes	Yes	DSL	Yes	Yes	Yes	Yes	Yes	Yes	Two year	Yes	Credit																
31	8773-HHUOZ	Female	0		No	Yes	17	Yes	No	DSL	No	No	No	No	Yes	Yes	Month-to-month	Yes	M																
32	3841-NFECK	Female	1		Yes	No	71	Yes	Yes	Fiber optic	Yes	Yes	Yes	Yes	No	No	Two year	Yes	Credit																
33	4929-XIHVV	Male	1		Yes	No	2	Yes	No	Fiber optic	No	No	Yes	No	Yes	Yes	Month-to-month	Yes	Credit																
34	6827-IEAUQ	Female	0		Yes	Yes	27	Yes	No	DSL	Yes	Yes	Yes	Yes	No	No	One year	No	M																
35	7310-EGVHZ	Male	0		No	No	1	Yes	No	No	No internet service	No internet service	No internet service	No internet service	No internet service	No internet service	Month-to-month	No	Bank tra																
36	3413-BMNZE	Male	1		No	No	1	Yes	No	DSL	No	No	No	No	No	No	Month-to-month	No	Bank tra																
37	6234-RAAPL	Female	0		Yes	Yes	72	Yes	Yes	Fiber optic	Yes	Yes	No	Yes	Yes	No	Two year	No	Bank tra																
38	6047-YHPVI	Male	0		No	No	5	Yes	No	Fiber optic	No	No	No	No	No	No	Month-to-month	Yes	Elec																
39	6572-ADKRS	Female	0		No	No	46	Yes	No	Fiber optic	No	No	Yes	No	No	No	Month-to-month	Yes	Credit																
40	5180-WJKOY	Male	0		No	No	34	Yes	Yes	Fiber optic	No	Yes	Yes	No	Yes	Yes	Month-to-month	Yes	Elec																
41	8168-UQWWF	Female	0		No	No	11	Yes	Yes	Fiber optic	No	No	Yes	No	Yes	Yes	Month-to-month	Yes	Bank tra																
42	8865-TNMDX	Male	0		Yes	Yes	10	Yes	No	DSL	No	Yes	No	No	No	No	One year	No	M																
43	9489-DEDVP	Female	0		Yes	Yes	70	Yes	Yes	DSL	Yes	Yes	No	No	Yes	No	Two year	Yes	Credit																
44	9867-JCZSP	Female	0		Yes	Yes	17	Yes	No	No	No internet service	No internet service	No internet service	No internet service	No internet service	No internet service	One year	No	M																
45	4671-VJLCL	Female	0		No	No	63	Yes	Yes	DSL	Yes	Yes	Yes	Yes	Yes	No	Two year	Yes	Credit																
46	4080-IARD	Female	0		Yes	No	13	Yes	Yes	DSL	Yes	Yes	No	Yes	Yes	No	Month-to-month	Yes	Elec																
47	3714-NTNFO	Female	0		No	No	49	Yes	Yes	Fiber optic	No	No	No	No	No	Yes	Month-to-month	Yes	Elec																
48	5948-UZLF	Male	0		No	No	7	Yes	No	DSL	No	Yes	No	No	No	No	Month-to-month	No	M																
49	7760-OYPDY	Female	0		No	No	2	Yes	No	Fiber optic	No	No	No	No	Yes	No	Month-to-month	Yes	Elec																



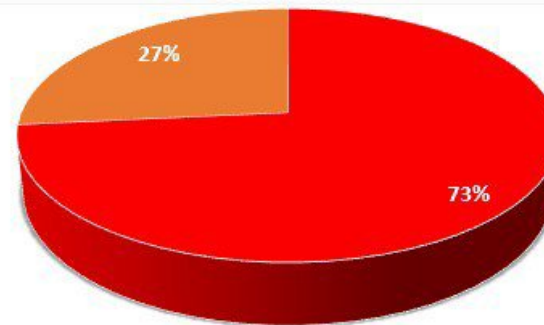


CHURN ANALYSIS VISUALIZATION

Row Labels	Count of customer_ID
No	5174
Yes	1869
Grand Total	7043

Count of customer_ID

CUSTOMERS ANALYSIS



Churn ▾

■ No

■ Yes

RISK CATEGORY VISUALIZATION

Row Labels	Count of customer_ID
High Risk	826
Low Risk	3833
Medium Risk	2384
Grand Total	7043



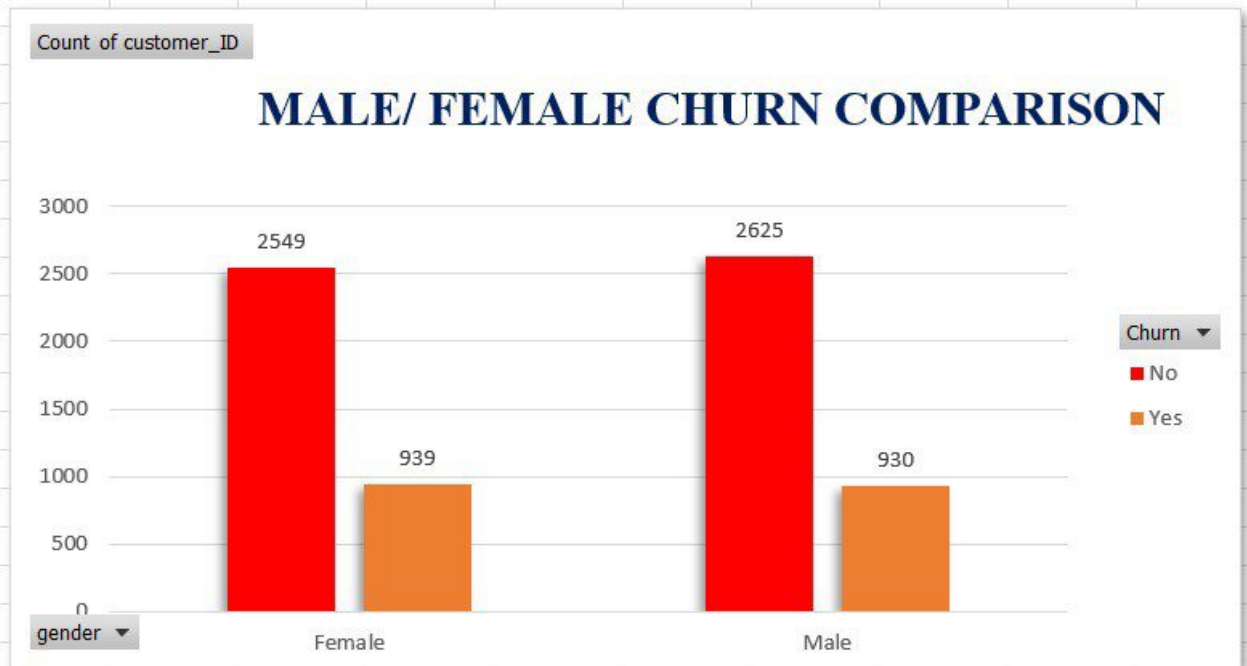
CUSTOMER VS PURPOSE

Count of customer_ID			
Row Labels	Column Labels		
	No	Yes	Grand Total
Month-to-month	2220	1655	3875
One year	1307	166	1473
Two year	1647	48	1695
Grand Total	5174	1869	7043



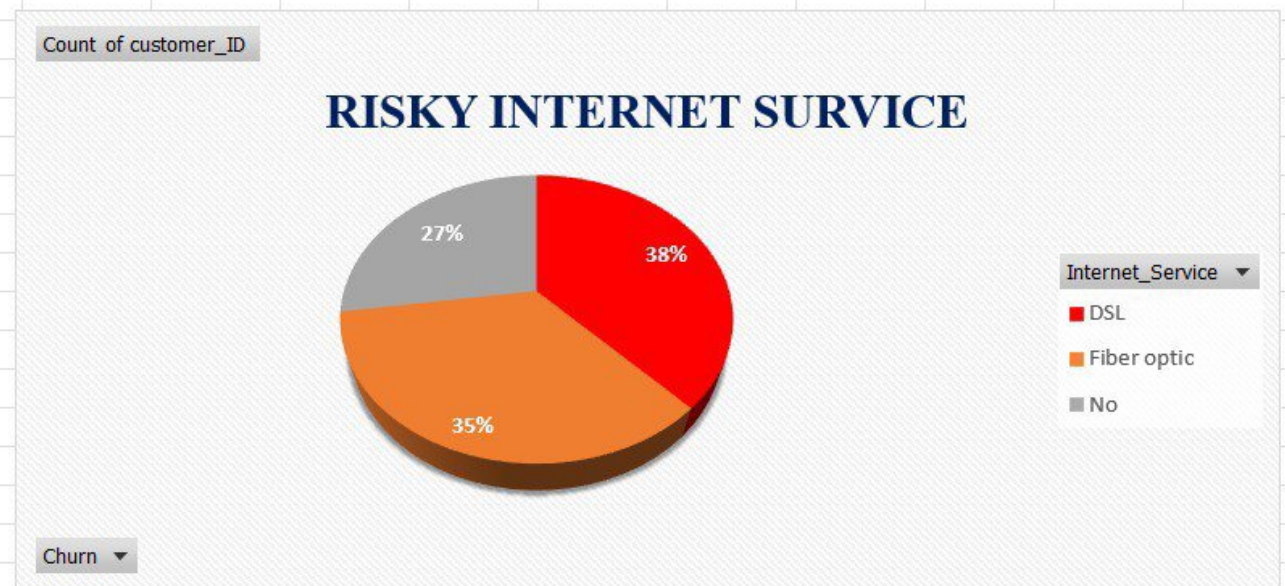
GENDER VS CHURN

Count of customer_ID	Column Labels		
Row Labels	No	Yes	Grand Total
Female	2549	939	3488
Male	2625	930	3555
Grand Total	5174	1869	7043



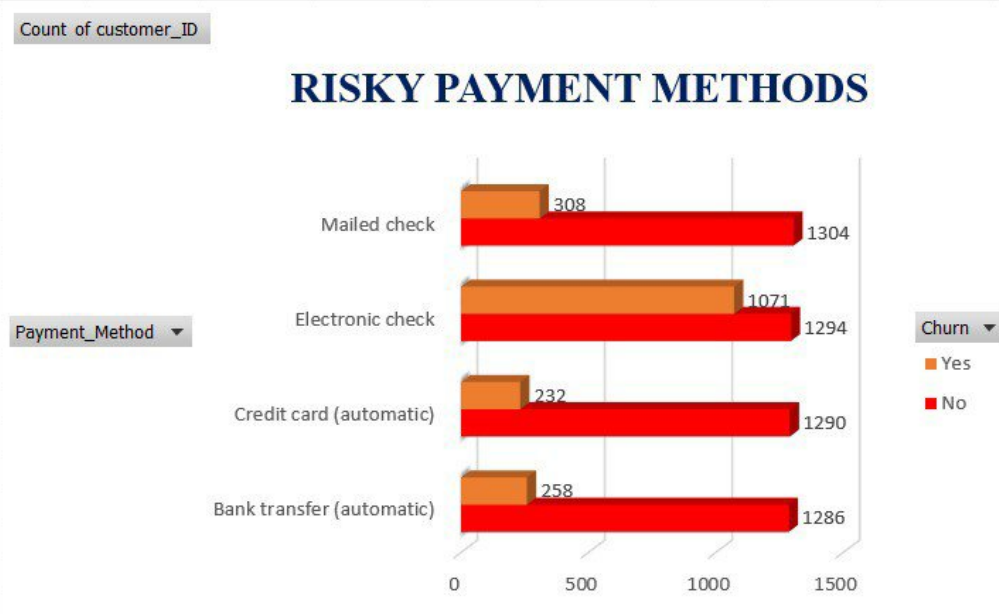
INTERNET SERVICE ANALYSIS

Count of customer_ID	Column Labels		
Row Labels	No	Yes	Grand Total
DSL	1962	459	2421
Fiber optic	1799	1297	3096
No	1413	113	1526
Grand Total	5174	1869	7043



PAYMENT METHOD ANALYSIS

Count of customer_ID	Column Labels		
Row Labels	No	Yes	Grand Total
Bank transfer (automatic)	1286	258	1544
Credit card (automatic)	1290	232	1522
Electronic check	1294	1071	2365
Mailed check	1304	308	1612
Grand Total	5174	1869	7043



TECH SUPPORT ANALYSIS

Count of customer_ID			
Column Labels			
Row Labels	No	Yes	Grand Total
No	2027	1446	3473
No internet service	1413	113	1526
Yes	1734	310	2044
Grand Total	5174	1869	7043

CHECK SUPPORT IMACT

Count of customer_ID

Tech_Support

Churn

No

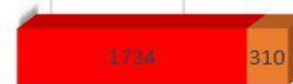
Yes

Yes

No

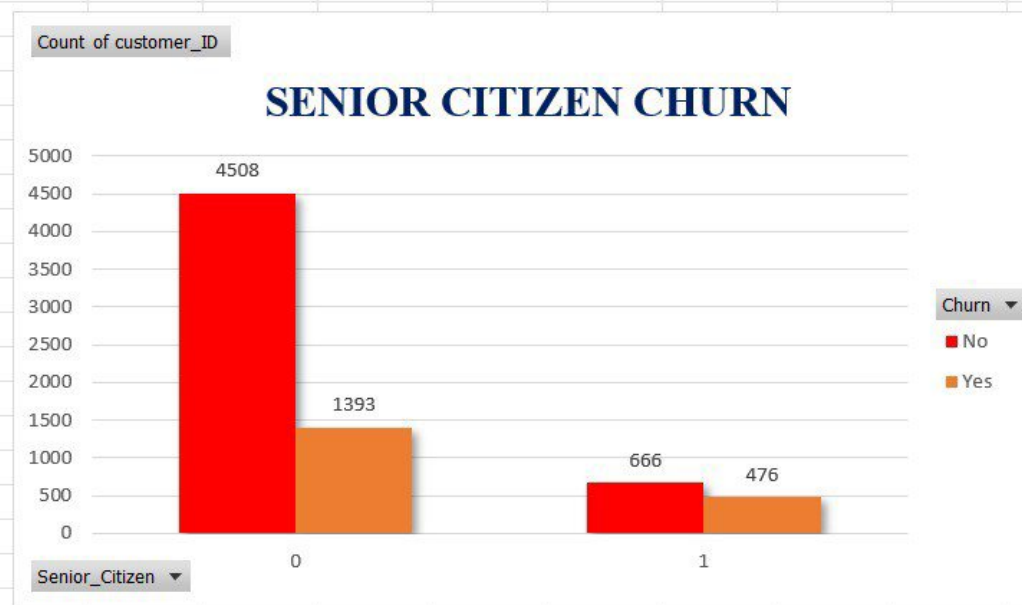
No internet service

0 1000 2000 3000 4000



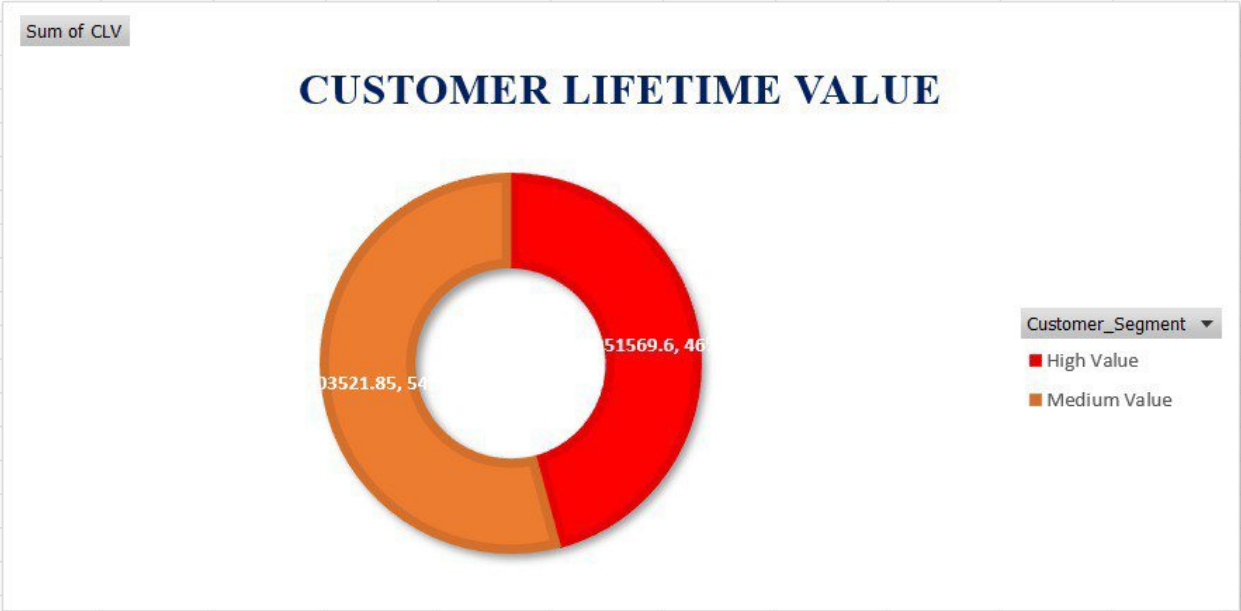
SENIOR CITIZEN VISUAL

Count of customer_ID			
Column Labels			
Row Labels	No	Yes	Grand Total
0	4508	1393	5901
1	666	476	1142
Grand Total	5174	1869	7043



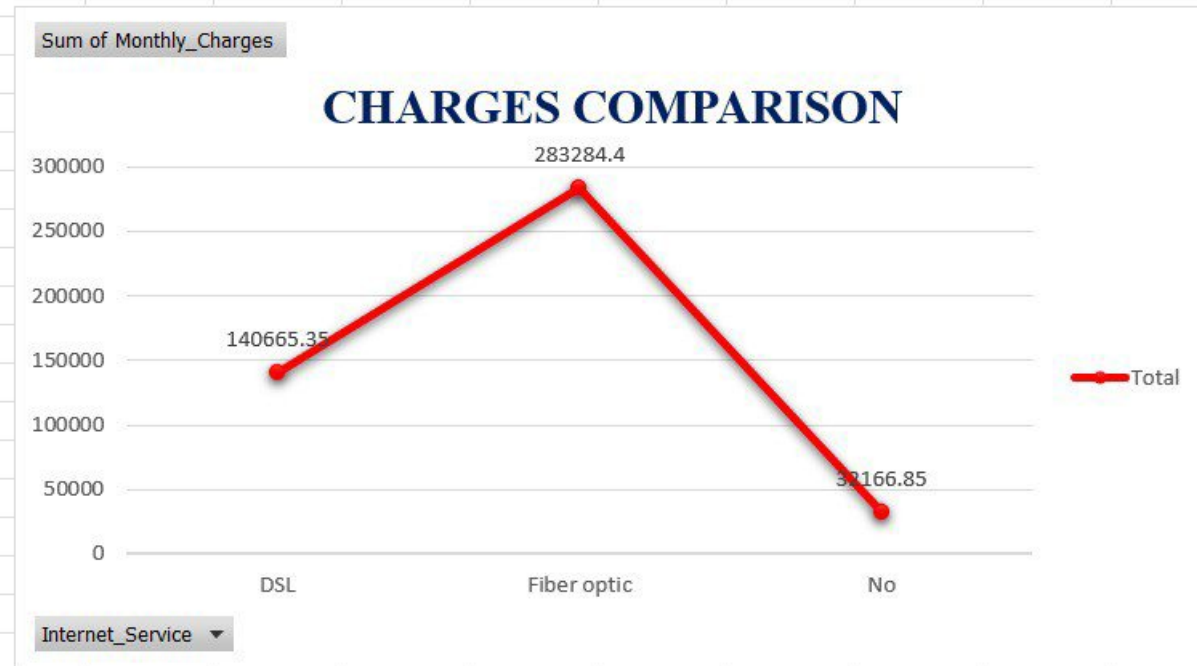
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
1																			
2																			
3																			
4																			
5																			
6																			
7																			
8																			
9																			
10																			
11																			
12	Row Labels	Sum of CLV																	
13	High Value	7351569.6																	
14	Medium Value	8703521.85																	
15	Grand Total	16055091.45																	
16																			
17																			
18																			
19																			
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32																			
33																			
34																			
35																			

CLV SEGMENT LIFETIME VALUE VISUAL



MONTHLY CHARGES VISUAL

Row Labels	Sum of Monthly_Charges
DSL	140665.35
Fiber optic	283284.4
No	32166.85
Grand Total	456116.6

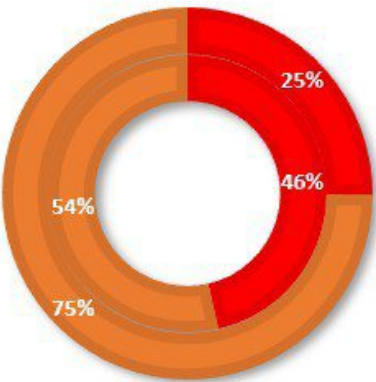


PAPERLESS BILLING VISUAL

Count of customer_ID			
Column Labels			
Row Labels	No	Yes	Grand Total
No	2403	469	2872
Yes	2771	1400	4171
Grand Total	5174	1869	7043

Count of customer_ID

BILLING IMPACT ON CHURN



Paperless_Billing

No

Yes

Churn

ONLINE SECURITY VISUAL

Count of customer_ID			
Column Labels			
Row Labels	No	Yes	Grand Total
No	2037	1461	3498
No internet service	1413	113	1526
Yes	1724	295	2019
Grand Total	5174	1869	7043

Count of customer_ID

SECURITY IMPACT ON CHURN

Online_Security

No internet service

Yes

No

Churn

No

Yes

